Problem

Let $n \geq m \geq 1$ be integers. Prove that

$$\frac{\gcd(m,n)}{n}\binom{n}{m}$$

is always an integer.

Source: **PUTNAM 2000**

§1 Harnessing Bézout's Identity

At first glance, the factor

$$\frac{\gcd(m,n)}{n}$$

appears troublesome, since dividing by n need not preserve integrality. The key observation is that Bézout's identity allows us to rewrite the greatest common divisor in a form that interacts naturally with the binomial coefficient.

By Bézout's Lemma, there exist integers x and y such that

$$\gcd(m,n) = mx + ny.$$

Substituting this expression into the quantity of interest gives

$$\frac{\gcd(m,n)}{n}\binom{n}{m} = \frac{mx+ny}{n}\binom{n}{m}.$$

Expanding,

$$\frac{\gcd(m,n)}{n}\binom{n}{m} = x \cdot \frac{m}{n}\binom{n}{m} + y\binom{n}{m}.$$

The second term is obviously an integer, since both y and $\binom{n}{m}$ are integers.

Thus it remains only to show that

$$\frac{m}{n}\binom{n}{m}$$

is an integer.

§2 The Binomial Coefficient Absorbs the Fraction

Using the factorial formula for binomial coefficients,

$$\frac{m}{n}\binom{n}{m} = \frac{m}{n} \cdot \frac{n!}{m!(n-m)!}.$$

Since

$$n! = n(n-1)!, \qquad m! = m(m-1)!,$$

we obtain

$$\begin{aligned} \frac{m}{n}\binom{n}{m} &= \frac{m}{n} \cdot \frac{n(n-1)!}{m(m-1)!(n-m)!} \\ &= \frac{(n-1)!}{(m-1)!(n-m)!}. \end{aligned}$$

But the final expression is exactly

$$\binom{n-1}{m-1},$$

which is an integer.

Therefore

$$x \cdot \frac{m}{n}\binom{n}{m} = x\binom{n-1}{m-1} \in \mathbb{Z}.$$

Both terms in the decomposition are integers, and hence their sum is an integer.

§3 The Final Conclusion

We have shown that

$$\frac{\gcd(m,n)}{n}\binom{n}{m} = x\binom{n-1}{m-1} + y\binom{n}{m},$$

where each term on the right-hand side is an integer.

Consequently,

$$\boxed{\frac{\gcd(m,n)}{n}\binom{n}{m} \in \mathbb{Z}}$$

for every pair of integers

$$n \geq m \geq 1.$$

This completes the proof.